

Michael Tang

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EDUCATION

Carnegie Mellon University

M.S. in Electrical & Computer Engineering (Deferred Spring 2027 Admission)

Pittsburgh, PA

Expected May 2028

University of California, Santa Barbara

B.S. in Computer Engineering; Dean's Honors, College of Engineering

Santa Barbara, CA

June 2026

EXPERIENCE

UCSB Novel Electronic Device & Circuits Lab

Santa Barbara, CA

VLSI Undergraduate Researcher, Hardware Accelerator VLSI

Oct 2025 – Aug 2026

- Contributed to tapeout of a 2 mm² Rocket + RoCC hardware accelerator for SAT-solving workloads, including floorplanning, power delivery, timing closure, and signoff in TSMC 28nm.
- Developed Python/TCL automation for Cadence Innovus power-grid generation, reducing power-strap spacing violations by 90%.
- Executed DRC and LVS signoff using Calibre and debugging implementation violations across logic and SRAM standard-cell regions prior to tapeout.

UCSB ArchLab

Santa Barbara, CA

Microarchitecture Undergraduate Researcher, CPU Microarchitecture

Sep 2025 – Jun 2026

- Contributed to an open-source RV64GC superscalar out-of-order core in SystemVerilog, focusing on branch prediction microarchitecture.
- Designing and validating a gshare branch predictor using cocotb to improve front-end performance.

UCSB Power Electronics and Microsystems Lab

Santa Barbara, CA

Undergraduate Student Researcher, Power IC Design

Apr 2025 – Oct 2025

- Assisted design and verification of a current-balanced integrated dual-inductor hybrid Dickson converter, reducing converter volume by 38%.
- Analyzed switch DC power/current versus DIH duty cycle to improve efficiency relative to a 3:1 integrated DIH Dickson converter baseline (Nguyen, Han, Pham-Nguyen, Le; IEEE APCCAS 2022).
- Verified circuit functionality and specifications using Cadence Virtuoso and LTSpice.

PROJECTS

ML-Based Fine-Grained RTL Timing Estimation | *Python, Yosys, OpenSTA, ML for EDA*

Winter 2026

- Developed an ML pipeline for register-level timing prediction using synthesis and timing data extracted from Yosys and OpenSTA.
- Engineered LambdaMART ranking models achieving 78.2% COVR in identifying the top 5% most timing-critical registers across 20 synthesized RTL designs.
- Built XGBoost regression models predicting global Worst Negative Slack (WNS) within 7.5% error while developing automated feature-extraction pipelines for large-scale timing datasets.

Optimized Chip Clock Tree in IBM 130nm | *HSpice, Digital IC Design*

Fall 2025

- Designed a clock distribution tree for a 5 mm² chip powering 100,000 flip-flops.
- Optimized clock jitter and rise-time to within 100 ps via buffer sizing, buffer spacing, and metal layer selection.
- Performed power analysis across 1 mm² power domains and validated performance using 1,000 Monte Carlo simulations (5th–95th percentiles).

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, VHDL, Python, Chisel, C/C++, TCL, Bash

EDA: Innovus, Virtuoso, Calibre, OpenSTA, Yosys, HSpice, LTSpice, ModelSim, Vivado

ASIC: Digital Design, RTL Synthesis, FPGA Prototyping, Place and Route, Power Analysis, Static Timing Analysis, Clock Tree Synthesis, cocotb